Blockchain
One

Overview

- BTC
- Bitcoin as a Currency
- The Components of the System
- Some of the Processes – Mining / Consensus
- The Software Protocol
- The History
- A Transaction Lifecycle
- Wallets

BTC / Bitcoin

- The 1st Blockchain and Crypto Currency
- Created / Appeared in 2009

- The Father of Crypto
- Speculated to be Created by Satoshi Nakamoto
- Wiki Link

- All other cryptocurrencies are considered as Alt-Coins


BANK
BTCMANAGER

Satoshi Nakamoto

- Wiki Link
- PsudoName – Not Real Name
- Reporters Did approach this individual but he denied ownership
- There are possible individuals who may be responsible
    - Hal Finey (Cryptographic Pioneer)
    - Nick Szabo (Smart Contracts)

Satoshi Nakamoto

- Alternative Theories Suggest :

- Some do Know , but the majority do not and may never know.

In the Beginning

- The First Transaction Took Place in 2009.
- Jan 3, 2009 6:15 PMUTC
- The value of BTC was : $0 and the amount was 50btc
- If that transaction took place in Nov 2021 at the height of BTC value, the transaction would be worth 50 X 65,000$
- The Genesis Block

- Entire Price History

What Else Happened at the start ?

- [P2PFoundation]
- [The Post]
- [The whitepaper]
  - More on WPs and ICOs Later

Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshin@gmx.com
www.bitcoin.org

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

1. Introduction

Commerce on the Internet has come to rely almost exclusively on financial institutions serving as trusted third parties to process electronic payments. While the system works well enough for most transactions, it still suffers from the inherent weaknesses of the trust based model. Completely non-reversible transactions are not really possible, since financial institutions cannot avoid mediating disputes. The cost of mediation increases transaction costs, limiting the minimum practical transaction size and cutting off the possibility for small casual transactions, and there is a broader cost in the loss of ability to make non-reversible payments for non-reversible services. With the possibility of reversal, the need for trust spreads. Merchants must be wary of their customers, hassling them for more information than they would otherwise need. A certain percentage of fraud is accepted as unavoidable. These costs and payment uncertainties can be avoided in person by using physical currency, but no mechanism exists to make payments over a communications channel without a trusted party.

What is needed is an electronic payment system based on cryptographic proof instead of trust, allowing any two willing parties to transact directly with each other without the need for a trusted third party. Transactions that are computationally impractical to reverse would protect sellers from fraud, and routine escrow mechanisms could easily be implemented to protect buyers. In this paper, we propose a solution to the double-spending problem using a peer-to-peer distributed timestamp server to generate computational proof of the chronological order of transactions. The system is secure as long as honest nodes collectively control more CPU power than any cooperating group of attacker nodes.

Bitcoin open source implementation of P2P currency

Posted by Satoshi Nakamoto on February 11, 2009 at 22:27

View Discussions

I've developed a new open source P2P e-cash system called Bitcoin. It's completely decentralized, with no central server or trusted parties, because everything is based on crypto proof instead of trust. Give it a try, or take a look at the screenshots and design paper:

Download Bitcoin v0.1 at http://www.bitcoin.org

The root problem with conventional currency is all the trust that's required to make it work. The central bank must be trusted not to debase the currency, but the history of fiat currencies is full of breaches of that trust. Banks must be trusted to hold our money and transfer it electronically, but they lend it out in waves of credit bubbles with barely a fraction in reserve. We have to trust them with our privacy, trust them not to let identity thieves drain our accounts. Their massive overhead costs make micropayments impossible.

A generation ago, multi-user time-sharing computer systems had a similar problem. Before strong encryption, users had to rely on password protection to secure their files, placing trust in the system administrator to keep their information private. Privacy could always be overridden by the admin based on his judgment call weighing the principle of privacy against other concerns, or at the behest of his superiors. Then strong encryption became available to the masses, and trust was no longer required. Data could be secured in a way that was physically impossible for others to access, no matter for what reason, no matter how good the excuse, no matter what.

It's time we had the same thing for money. With e-currency based on cryptographic proof, without the need to trust a third party middleman, money can be secure and transactions effortless.

One of the fundamental building blocks for such a system is digital signatures. A digital coin contains the public key of its owner. To transfer it, the owner signs the coin together with the public key of the next owner. Anyone can check the signatures to verify the chain of ownership. It works well to secure ownership, but leaves one big problem unsolved: double-spending. Any owner could try to re-spend an already spent coin by signing it again to another owner. The usual solution is for a trusted company with a central database to check for double-spending, but that just gets back to the trust model. In its central position, the company can override the users, and the fees needed to support the company make micropayments impractical.

Bitcoin's solution is to use a peer-to-peer network to check for double-spending. In a nutshell, the network works like a distributed timestamp server, stamping the first transaction to spend a coin. It takes advantage of the nature of information being easy to spread but hard to stifle. For details on how it works, see the design paper at http://www.bitcoin.org/bitcoin.pdf

The result is a distributed system with no single point of failure. Users hold the crypto keys to their own money and transact directly with each other, with the help of the P2P network to check for double-spending.

Satoshi Nakamoto
http://www.bitcoin.org

The Humour & Who Could Have Known ?

- Bitcoin Pizza Day – May 22nd
- In May, 2010 Laszlo Hanyecz set out to buy two large pizzas with bitcoin.
- The Post
- How Much was it worth Then and how Much is it worth now? Use BC

I'll pay 10,000 bitcoins for a couple of pizzas.. like maybe 2 large ones so I have some left over for the next day. I like having left over pizza to nibble on later. You can make the pizza yourself and bring it to my house or order it for me from a delivery place, but what I'm aiming for is getting food delivered in exchange for bitcoins where I don't have to order or prepare it myself, kind of like ordering a 'breakfast platter' at a hotel or something, they just bring you something to eat and you're happy!

I like things like onions, peppers, sausage, mushrooms, tomatoes, pepperoni, etc.. just standard stuff no weird fish topping or anything like that. I also like regular cheese pizzas which may be cheaper to prepare or otherwise acquire.

If you're interested please let me know and we can work out a deal.

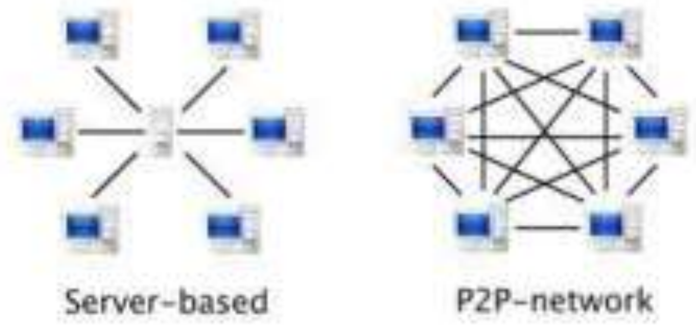
Thanks,
Laszlo

BC: 157fRrqAKrDyGHr1Bx3yDxeMv8Rh45aUet

https://blockchair.com/

Technical Infrastructure

- Peer to Peer (P2P)
- Decentralised
- Immutable
- Distributed, shared ledger of events



- What Applications do you think this tech would be Useful ?

A Ledger

- There exists some Issues with ledgers
- Susceptible to fraud
- Susceptible to human error
- Rely on intermediary to prevent double spend e.g. bank, paypal
- Full control out of the hands of the consumer
- In some cases the power of these

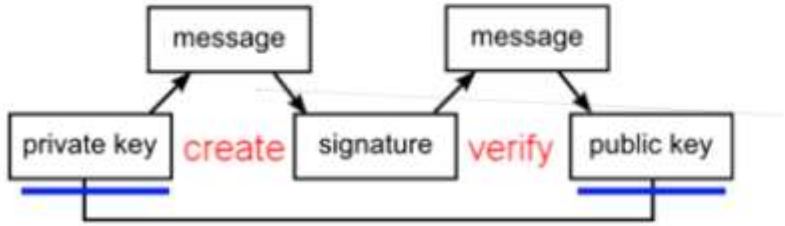
intermediaries can result in over-extensions of power:

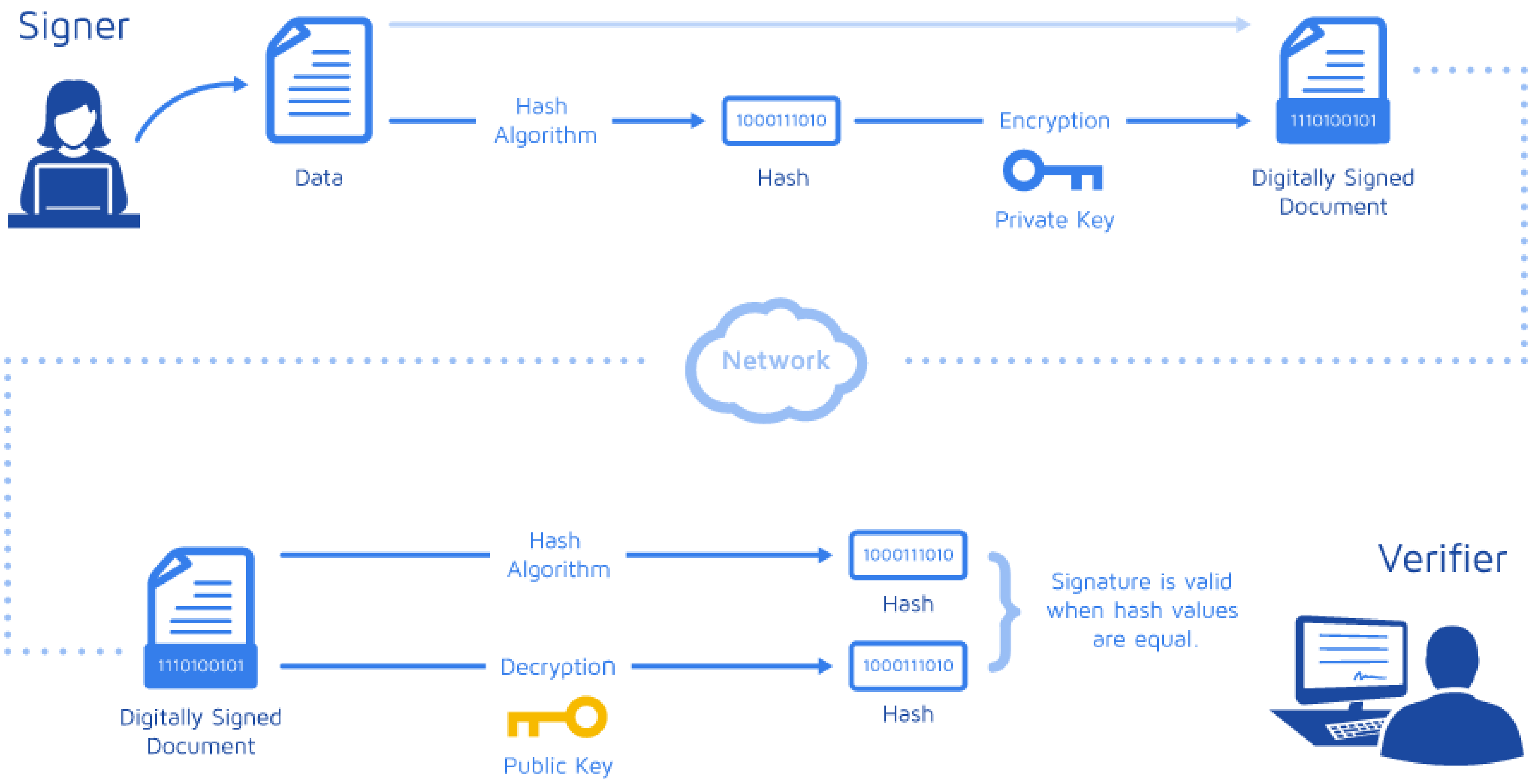
- Paypal block payments to Wikileaks etc

Cryptographic Beginnings

- BTC is a paradigm
- A collection of Technologies combined to provide a new tec / service
- Distributed Computing
- Peer to Peer
- Digital Signatures
- Consensus
- Hashing Algorithms

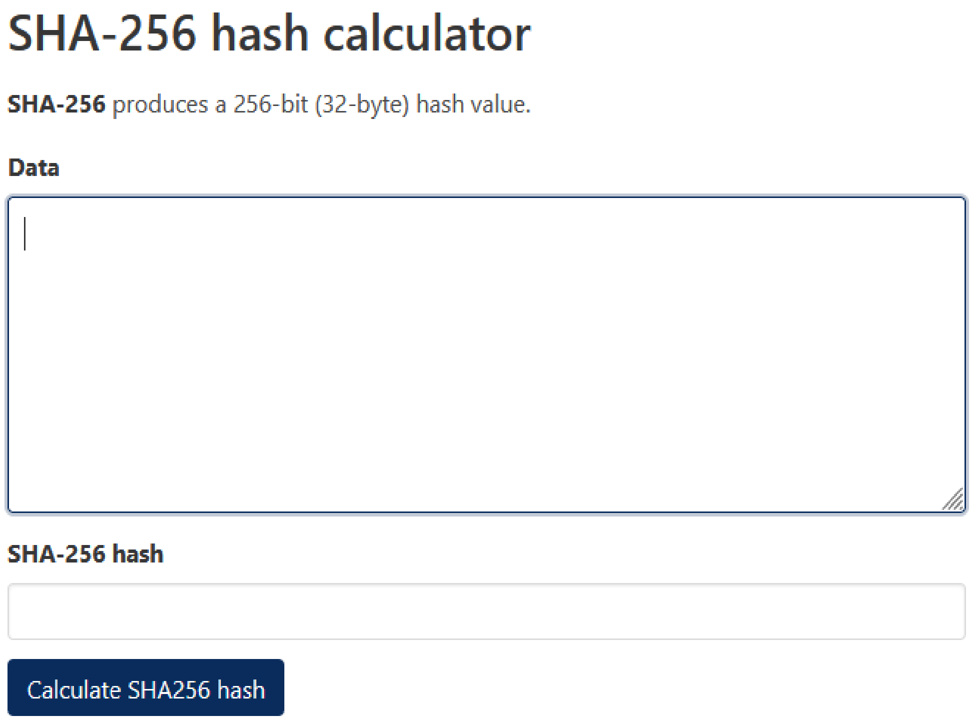
Digital Signature
Non Repudiation





Hashing Demo / BTC Functions Demo

- SHA-256
- “The SHA (Secure Hash Algorithm) is one of a number of cryptographic hash functions. A cryptographic hash is like a signature for a text or a data file. SHA-256 algorithm generates an almost-unique, fixed size 256-bit (32-byte) hash. Hash is a one way function – it cannot be decrypted back. This makes it suitable for password validation, challenge hash authentication, anti-tamper, digital signatures.”



https://andersbrownworth.com/blockchain/ - Fantastic Resource For Beginners to understand some of the workings

Hashing / SHA

- Hashing , given from cryptography is an irreplaceable feature within the BTC Network.
- It is used throughout
  - Verifying Transactions
  - Ensuring No Modifications / Immutability
  - Used to Perform Computational work
  - Those workings are then used for POW , proof of work
  - Demo of SHA-256 – SHA Calculator

Consensus "To Achieve a Consensus"

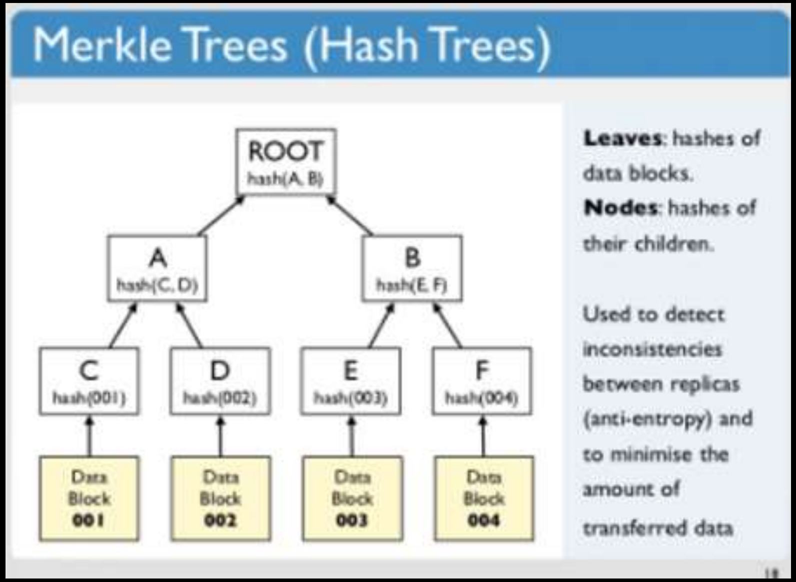
- BTC uses a POW consensus Mechanism
- Proof of Work
- There are several Variations of Consensus mechanisms across differing Blockchains.
- Ethereum recently converted to a POS , proof of stake.
- We will investigate some other Consensus Mechanisms as we progress.
- BTC First with POW

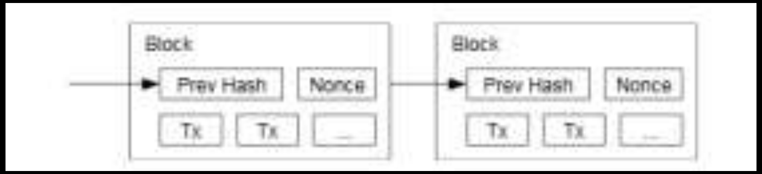
Proof Of Work

Hashcash was proposed in 1997 by Adam Back and described more formally in Back's 2002 paper "Hashcash - A Denial of Service Counter-Measure".

- **Proof of Work:**
- Based on Adam Back's Hashcash (2002) the PoW involves scanning for a value that when hashed, the hash begins with a number of zero bits.
- Miners (which we will discuss in more detail later) confirm proof of work by incrementing a nonce in the block until a value is found that gives the required zero bits to solve the block's hash.
- This PoW also solves the problem of determining representation in majority decision making.
- Proof-of-Work is essentially 'one-CPU-one-Vote'. The majority decision is represented by the longest chain, which has the greatest PoW efforts invested in it.
- To modify a past block, an attacker would have to redo the PoW of the block and all blocks after it and then catch up with and surpass the work of the honest nodes.
- The likelihood of this diminishes exponentially as subsequent blocks are added.

HashCash PDF


Merkle Trees (Hash Trees)
ROOT
hash(A, B)
A
hash(C, D)
B
hash(E, F)
C
hash(001)
D
hash(002)
E
hash(003)
F
hash(004)
Data Block 001
Data Block 002
Data Block 003
Data Block 004
Leaves: hashes of data blocks.
Nodes: hashes of their children.
Used to detect inconsistencies between replicas (anti-entropy) and to minimise the amount of transferred data


Block
Prev Hash
Nonce
Tx
Tx
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Block
Prev Hash
Nonce
Tx
Tx
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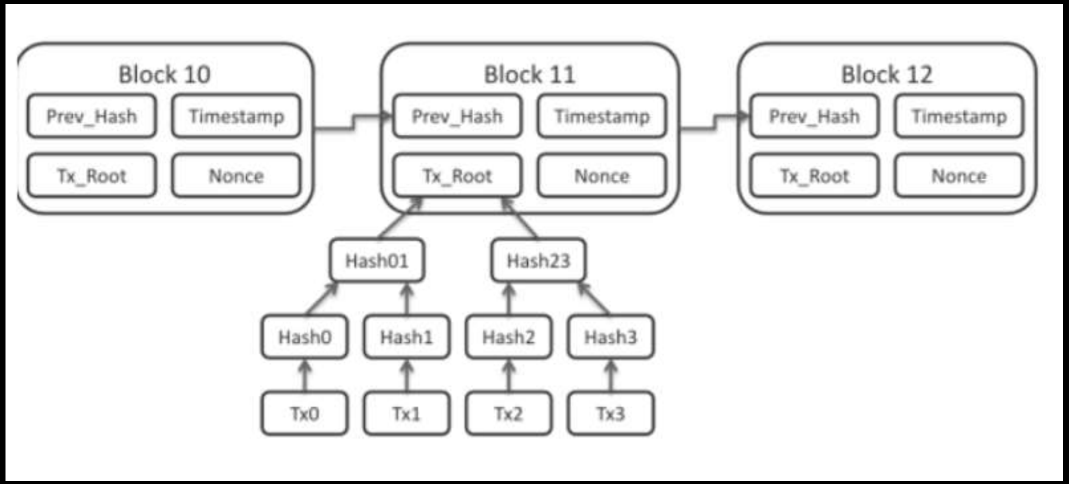
Block 10
Prev_Hash
Timestamp
Tx_Root
Nonce
Block 11
Prev_Hash
Timestamp
Tx_Root
Nonce
Block 12
Prev_Hash
Timestamp
Tx_Root
Nonce
Hash01
Hash23
Hash0
Hash1
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Hash3
Tx0
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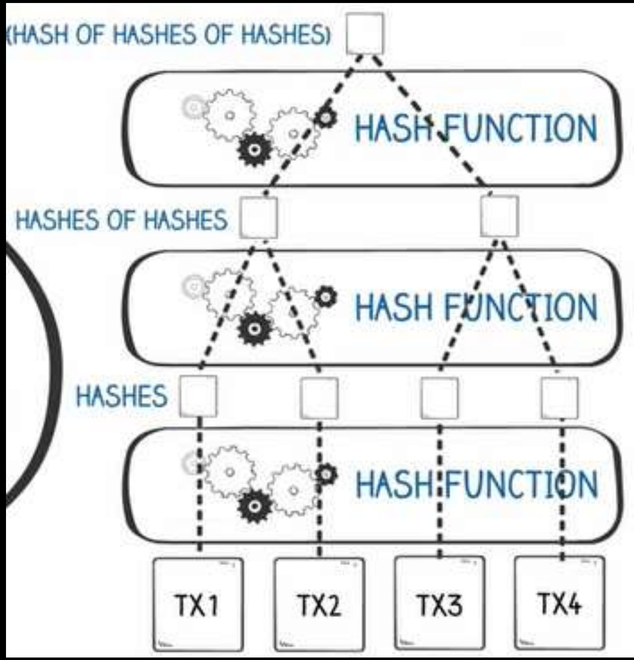
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TX1
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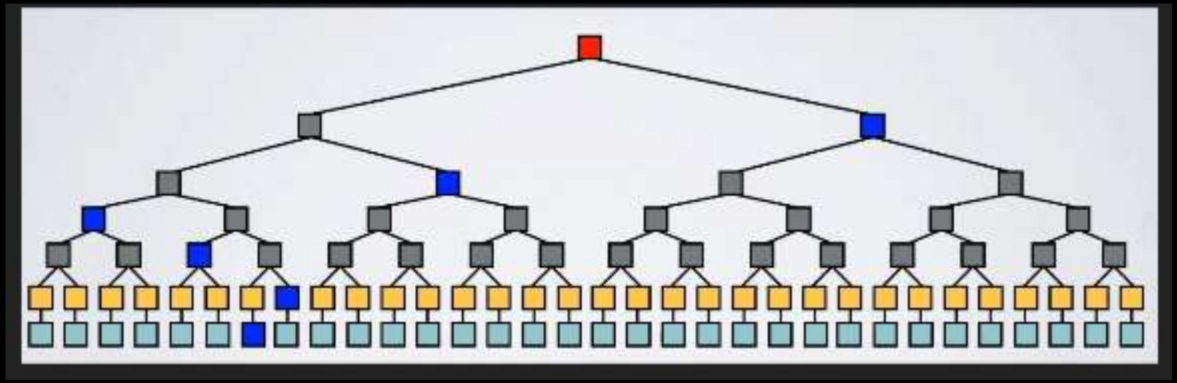
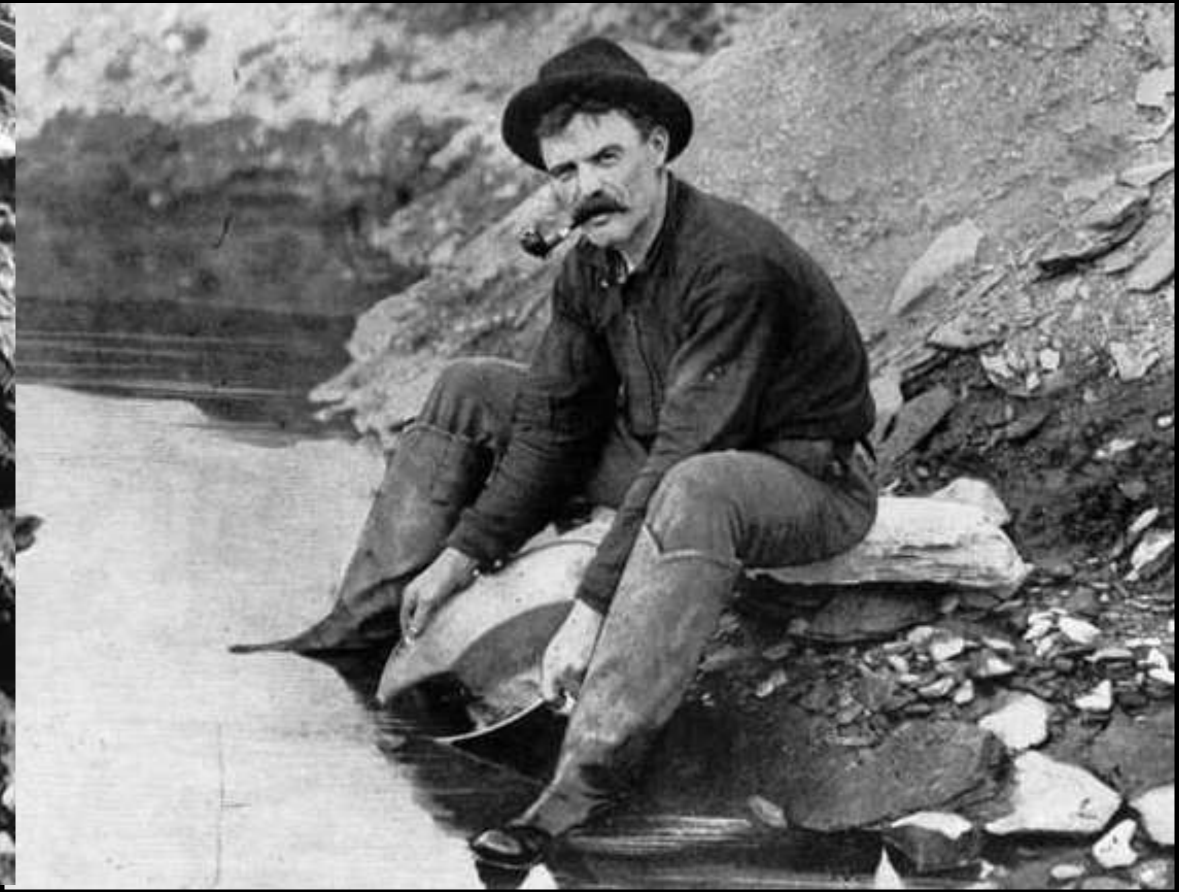


The Miners



The Miners

Mining

- This is the process by which new Bitcoin is added to the money supply. It also serves to protect the network against fraudulent transactions and double-spend and is based upon the key concepts discussed to this point.
- Essentially, miners provide processing power to the network in exchange for the opportunity to earn bitcoin as reward.
- Miners validate new transactions and record them on the blockchain, the global, distributed and shared ledger. Blocks are mined every 10 minutes, with a block containing all transactions since the last block was mined. Once the block is added to the blockchain the transactions are confirmed
- Miners receive rewards for mining blocks; new coins for creating a new block and transaction fees for the transactions within the block.
- The party who successfully mines the block is the party who first solves the Proof-of-Work algorithm

Difficulty & Rewards

- The Current Difficulty Level in BTC :
- https://www.coinwarz.com/mining/bitcoin/difficulty-chart

- 17 Leading Zeros
- https://en.bitcoin.it/wiki/Difficulty

- The Current Reward :
- The current bitcoin block reward is composed of **6.25 newly generated coins per block**.

- https://buybitcoinworldwide.com/halving/